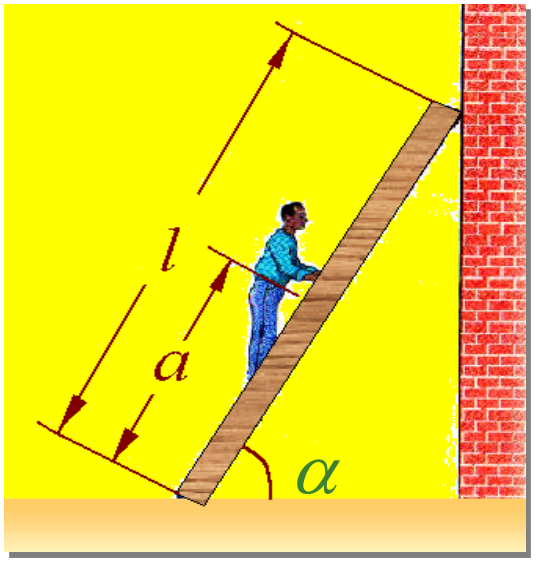
哈爾濱工業大學深圳研究生院
Harbin Institute of Technology Shenzhen Graduate School

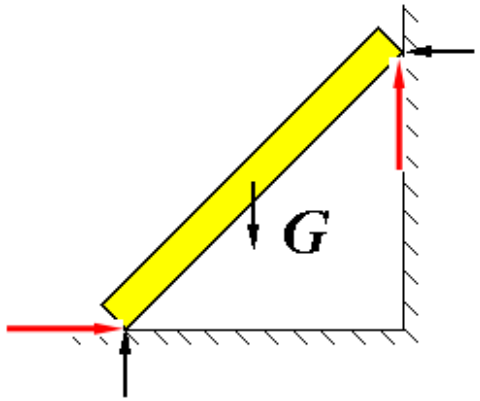
Chapter 4 Friction

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In previous chapters, we treated all contact surfaces as absolutely smooth, ignoring friction between objects. In reality, completely smooth surfaces do not exist; friction is generally present under normal circumstances.





equilibrium must consider friction

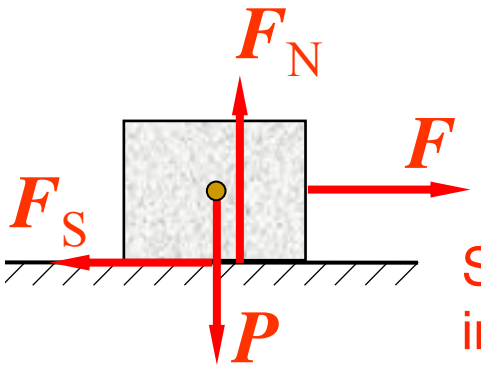
According to the motion of the contact surface, friction is divided into: Sliding friction, rolling friction

4.1 Sliding friction

I, Static sliding friction

1. Definition: When two objects in contact generate a relative **sliding trend**, the force generated by their contact surface to prevent the relative motion trend of the objects is called **static (sliding) frictional force**.
(It is the **tangential** constraint force exerted by the contact surface on an object)

2. Status:



① Static state: $F = F_s$, F_S increases with the increase of F, but there is a critical value

② Critical state: (Will but not slip) maximum static friction force
Static sliding friction coefficient

$$F_{max} = f_S \cdot F_N \quad (f_S —$$ Static sliding friction coefficient

So the way to increase friction is to:

① increase the normal pressure F_N

② increase the static friction coefficient f_S

3. Features:

Characteristics of static friction force:

- Magni-tude: $0 \le F_S \le F_{max}$ (equilibrium) satisfies $\sum F_x = 0$
- Direction: opposite to the sliding trend of the object - impending motion
- Law: $F_{max} = f_S \cdot F_N$

(f_s is only related to the material and surface condition, and is not related to the size of the contact area.)

II, Dynamic sliding friction (different from static sliding friction in that sliding occurs)

Characteristics of dynamic friction force:

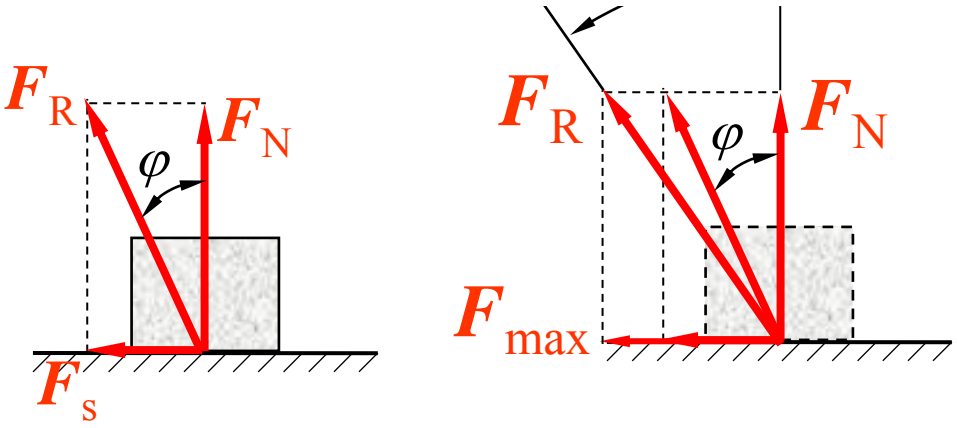
- Magni tude: $F_d = f \cdot F_N$ (No equilibrium range)
- Direction: opposite to the direction of object motion
- Law: $F_d = f \cdot F_N$

(*f* is only related to the material and surface condition, and is not related to the size of the contact area.)

4.2 Friction angle and self-locking phenomenon

I、 Friction angle

① Full constraint force, $\boldsymbol{F}_{\mathrm{R}}=\boldsymbol{F}_{\mathrm{N}}+\boldsymbol{F}_{\mathrm{S}}$, forms an angle φ with the common normal line of the contact surface. When the object is in a critical equilibrium state, i.e., when the static friction force reaches its maximum value F_{max}, the angle φ reaches its maximum value φ_f. The maximum angle φ_f. between the full constraint force and the normal is called the friction angle.



② Calculate $\tan\varphi_{\mathrm{f}}=\dfrac{F_{\max}}{F_{\mathrm{N}}}=\dfrac{f_{\mathrm{s}}F_{\mathrm{N}}}{F_{\mathrm{N}}}=f_{\mathrm{s}}$

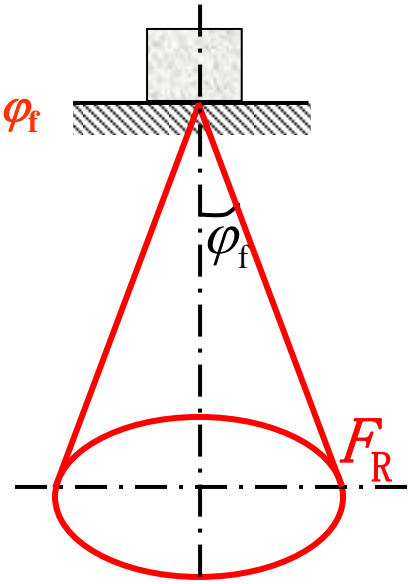
The tangent of the friction angle is equal to the static friction coefficient

When the sliding direction of the block changes, the orientation of the full constraint force line also changes accordingly;

At the critical state, the action line of F_R will draw a conical surface with contact point A as the vertex, called a friction cone.

If the coefficient of friction between the block and the supporting surface in any direction is the same, that is, the friction angle is equal, then the friction cone will be a cone with a vertex angle of $2\varphi_f$.

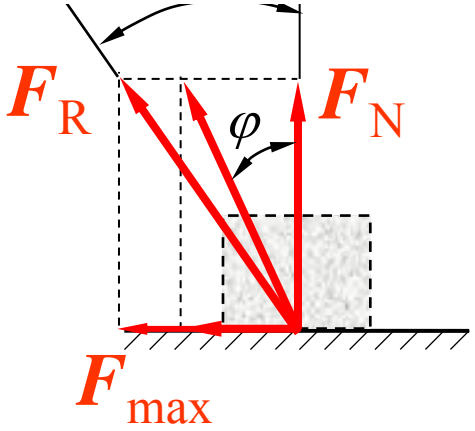
③ Friction cone: a cone with a vertex angle of $2\varphi_f$



II、 Self-locking phenomenon

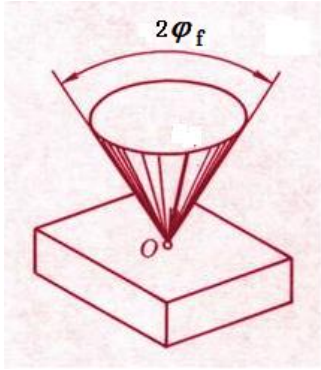
When the object is equilibrium, the static friction force may not necessarily reach its maximum value and can vary between zero and the maximum value F_{max}, so the angle φ between the full constraint force and the normal is also changing between zero and the friction angle φ_f,

$$0 \le \varphi \le \varphi_f$$



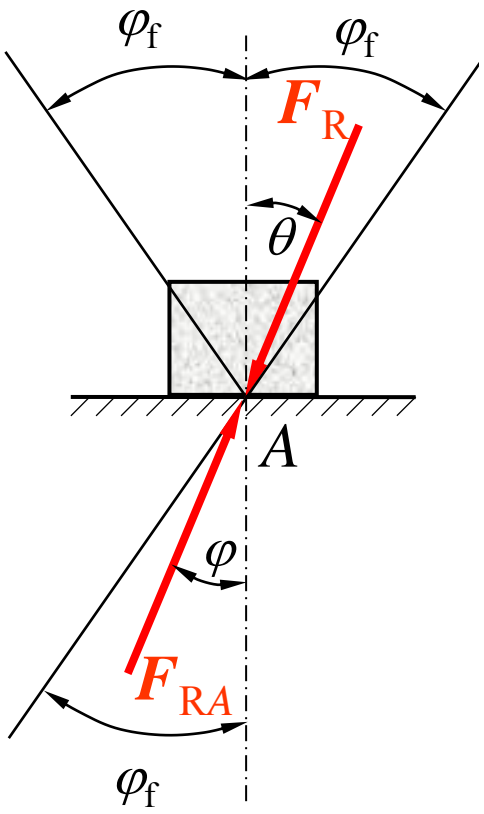
Since the static friction force cannot exceed the maximum value, the line of action of the full constrain force cannot exceed the friction angle, that is, the full constraint force must be within the friction angle.

Reverse the friction cone and determine if the active forces are within the reverse friction cone to determine if it is equilibrium.



1. If the line of action of the resultant force F_R of all active forces acting on the block is within the friction angle φ_f, then regardless of the magnitude of this force, the block must remain stationary. This phenomenon is called self-locking phenomenon.

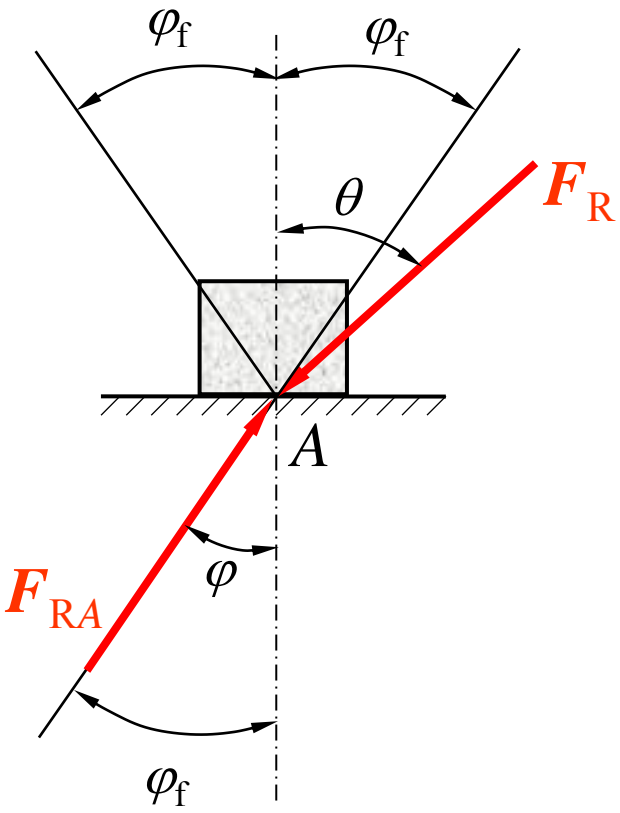
   Because in this case, the angle between the resultant force F_R of the active force and the normal line is less than $\theta<\varphi_f$, therefore, F_R and the full constraint force F_{RA} must satisfy the two force equilibrium condition



2. If the line of action of the resultant force F_R of the main force is outside the friction angle φ_f, then no matter how small this force is, the object will definitely slide.

Because in this case, $\theta > \varphi_f$, while $\varphi \leq \varphi_f$, the total restraining force F_{RA} of the supporting surface and the resultant force F_R of the active force cannot satisfy the two force equilibrium condition.

By applying this principle, we can try to avoid the occurrence of self-locking phenomenon.



4.3 The euilibrium with Consideration of friction

The equilibrium with Consideration of friction

- Determining whether force system is equilibrium
- Equilibrium Range Problem
- Critical equilibrium problem

I, Determining whether force system is equilibrium

(1) Assuming the object is in equilibrium, the frictional force F_S is taken as an unknown force and points towards a hypothetical direction

(2) Calculate F_S and F_N from the equilibrium equation. If F_S is negative, it points in the opposite direction of the hypothesis

(3) Calculate $F_{max} = f_S F_N$, and determine whether the system is equilibrium based on whether the inequality $| F_S | \leq F_{max}$ is satisfied. If the inequality is satisfied, it is balanced; otherwise, it is not equilibrium.

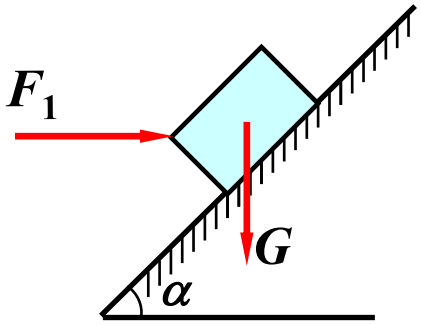
II、 The problem of finding the range of equilibrium (including force and geometric range):

1. Assuming an object is in a **critical equilibrium** and the frictional force reaches its **maximum** value F_{max}, its direction cannot be assumed, but should **be judged** based on the trend of object motion;
2. Supplementary equation $F_{max} = f_S F_N$, calculate the unknown quantity from the equilibrium equation;
3. Analyze the equilibrium range based on a certain critical equilibrium condition obtained.

III、 Critical equilibrium problem

1. Assuming an object is in **critical equilibrium** and the frictional force reaches its maximum value F_{max}, its direction cannot be assumed, but should be judged based on the trend of object motion;
2. Add the equation $F_{max} = f_S F_N$ and calculate the unknown variable from the equilibrium equation.

Example: On a fixed inclined surface with an inclination angle α greater than the friction angle φ_f, a heavy object *G* is placed. In order to maintain this object stationary on the inclined surface, a horizontal force F_1 is applied to the object. Try to determine the magnitude of this horizontal force.



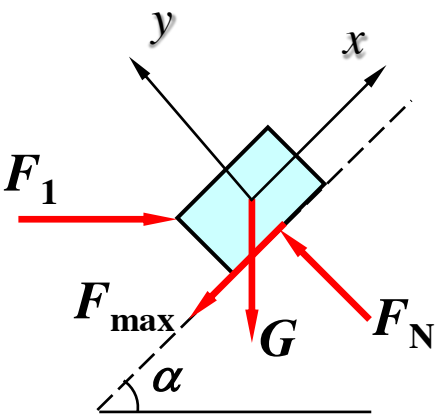
Solution: The block is the research object

1. Find the maximum F_{1max} of F_1, and analyze the force as shown in the figure.

Equilibrium equation:

$$\sum F_x = 0 \qquad F_1 \cos\alpha - F_{\max} - G\sin\alpha = 0$$

$$\sum F_y = 0 \qquad F_N - G\cos\alpha - F_1 \sin\alpha = 0$$



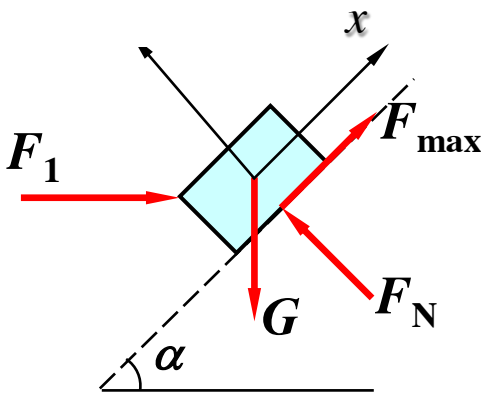
supplementary equation $F_{\max} = f_s F_N$

Simultaneous solution:

$$F_{1\max} = G\frac{\tan\alpha + \tan\varphi_f}{1 - \tan\varphi_f \tan\alpha} = G\tan(\alpha + \varphi_f)$$

2. Find the minimum F_{1min} of F_1, and analyze the force as shown in the figure.

Apply the same solution:



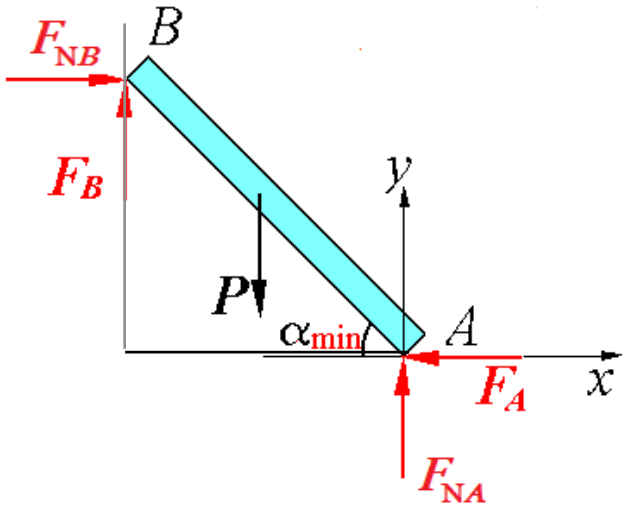
$$F_{1\min} = G\frac{\tan\alpha - \tan\varphi_{\mathrm{f}}}{1+\tan\varphi_{\mathrm{f}}\tan\alpha} = G\tan(\alpha-\varphi_{\mathrm{f}})$$

When equilibrium, F_1 should satisfy:

$$G\tan\left(\alpha-\varphi_{\mathrm{f}}\right) \le F_1 \le G\tan\left(\alpha+\varphi_{\mathrm{f}}\right)$$

[Example] If the length of the ladder is $AB=l$ and the weight is P, and the static friction coefficient f_s between the ladder and the wall and ground is 0.5, what is the value of α that the ladder can be in equilibrium?

Solution: Considering that the ladder has a downward trend in the critical equilibrium state, draw free body diagram.



$$\sum F_x = 0 \quad F_{NB} - F_A = 0 \qquad (1)$$

$$\sum F_y = 0 \quad F_{NA} + F_B - P = 0 \qquad (2)$$

$$F_A = f_s \cdot F_{NA} \qquad F_B = f_s \cdot F_{NB}$$

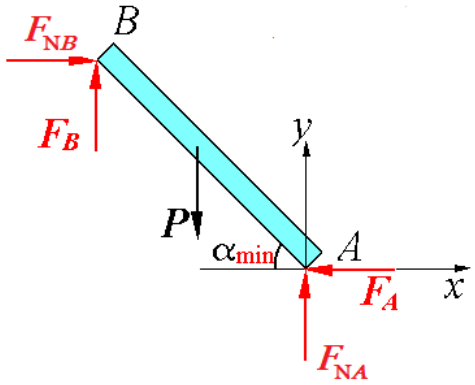
$$\sum M_A = 0 \quad P \cdot \frac{l}{2}\cos\alpha_{\min} - F_B \cdot l\cos\alpha_{\min} - F_{NB} \cdot l\sin\alpha_{\min} = 0 \qquad (3)$$

Solve $F_{NA}=\dfrac{P}{1+f_s^2}$, $F_{NB}=\dfrac{f_s P}{1+f_s^2}$, $F_B=\dfrac{f_s^2 P}{1+f_s^2}$ Solve（3）

Solve

$$\tan\alpha_{\min}=\frac{1-f_s^2}{2f_s}=\frac{1-\tan^2\varphi_f}{2\tan\varphi_f}=\operatorname{ctg}2\varphi_f=\tan\left(\frac{\pi}{2}-2\varphi_f\right)$$

$$\alpha_{\min}=\frac{\pi}{2}-2\varphi_f=36^0 87'$$



Note: α should be less than 90^0

So:

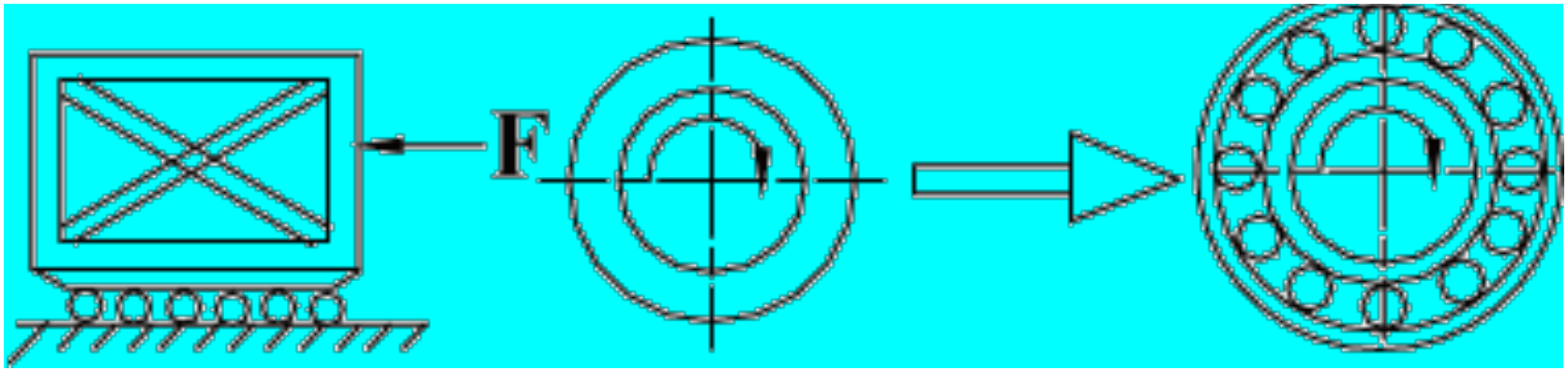
$$36^0 87' \le \alpha \le 90^0$$

4.4 Rolling friction

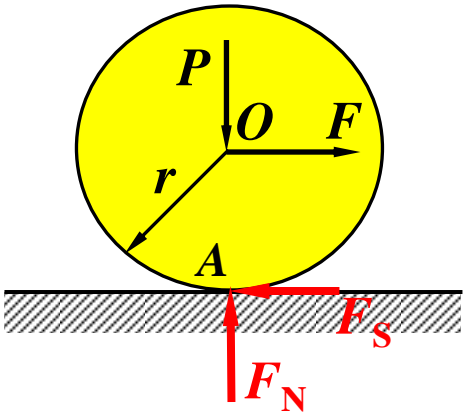
When carrying heavy objects, if a roller shaft is placed under the heavy object, it is much easier to push or pull than directly placing the heavy object on the ground.

This indicates that using the rolling of the roller shaft instead of sliding at the bottom of the box will result in less resistance.

Vehicles use wheels to "walk" and machines use rolling bearings to reduce frictional resistance.



It is known from practice that the roller reduce its sliding effort. The force analysis in the following figure shows a problem, that is, if the object is balanced, it cannot fully satisfy the equilibrium equation.



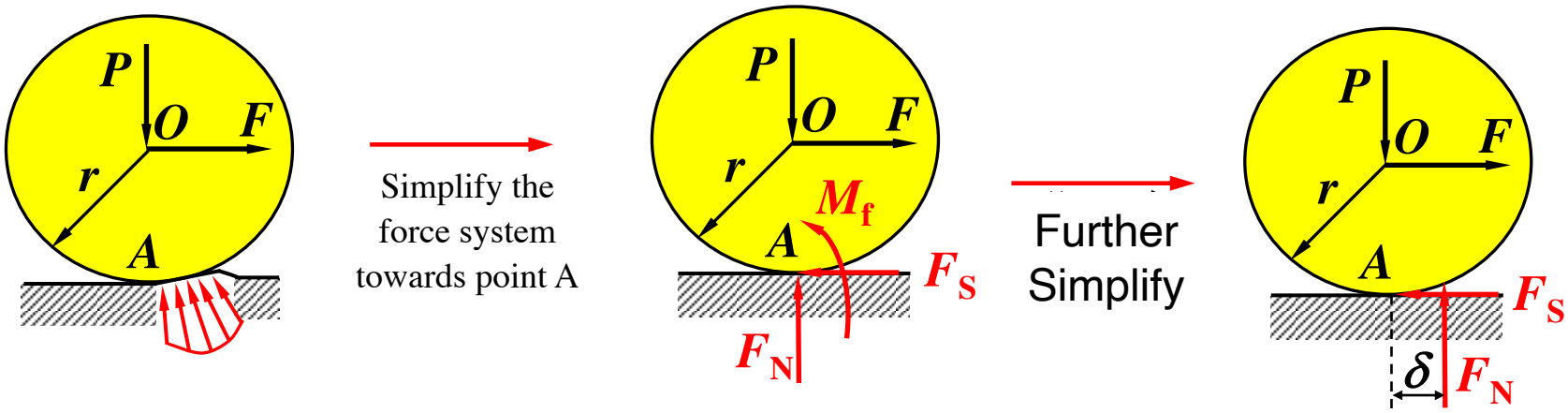
$$\sum F_x = 0 \quad F - F_S = 0$$

$$\sum F_y = 0 \quad F_N - P = 0$$

$$\sum M_A(F) = 0 \quad F \cdot r = 0 \text{ not satisfied}$$

F and F_S form an active force couple that causes the wheels to roll, but the wheels can balance.

The reason for this phenomenon is that, The actual contact surface is not a rigid body, they Under the action of force, some deformation will occur



The rolling friction couple M_f is balanced with the active couple (F, F_S)

rolling friction

① The rolling resistance couple M_f increases with the increase of the active force couple (F, F_S);

② $0 \leqq M_f \leqq M_{max}$, The friction moment has a range

③ M_{max} Independent of roller radius

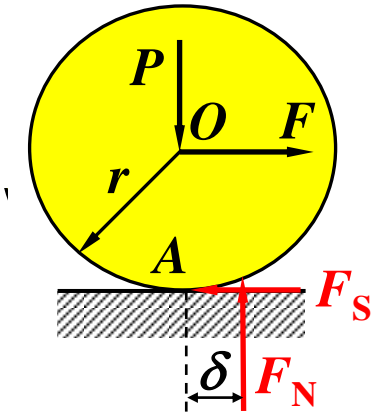
④ Rolling friction law $M_{max} = \delta F_N$, δ Rolling friction coefficient

Note for Rolling friction coefficient

① There is a length dimension, usually measured in cm, mm

② Related to the hardness and temperature of the materials used for the rollers and bearing surfaces

③ δ physical meaning is illustrated in the figure:



From the figure, it can be seen that the force arm of the rolling resistance pair M_f is exactly (rolling resistance coefficient), so has a length dimension. Due to the small rolling resistance coefficient, in most cases in engineering, rolling resistance is negligible, that is, rolling friction is ignored.

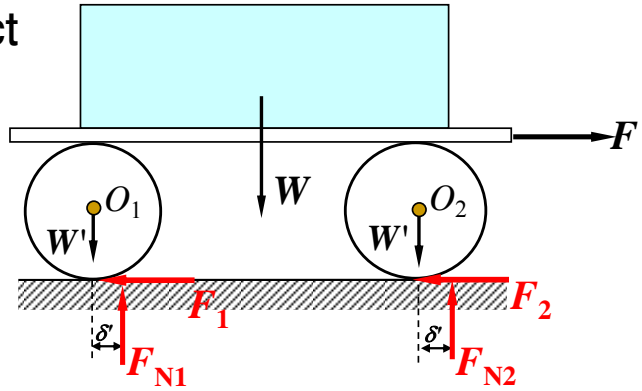
Active force required for rolling $F_{滚}=\dfrac{M_{\max}}{r}=\dfrac{\delta F_N}{r}=\dfrac{\delta}{r}P$

Active force required for sliding $F_{滑}=F_{\max}=f_s F_N=f_s P$

Generally speaking, $\dfrac{\delta}{r}\ll f_s$ scrolling is easier than sliding

Example: When carrying heavy objects, rollers are often placed underneath, as shown in the picture. Assuming the weight of the heavy object is W, the weight of the rolling wood is W', the radius is r, the rolling resistance coefficient between the rolling wood and the heavy object is δ, and the rolling resistance coefficient between the rolling wood and the ground is δ'. Find the magnitude of the horizontal force F when it is about to be pulled.

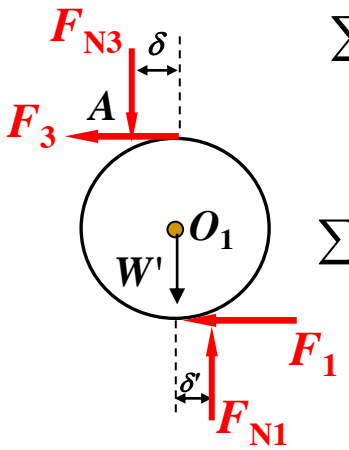
Solution: 1. Take the whole as the research object



$$\sum F_x = 0 \quad F - F_1 - F_2 = 0 \qquad (1)$$

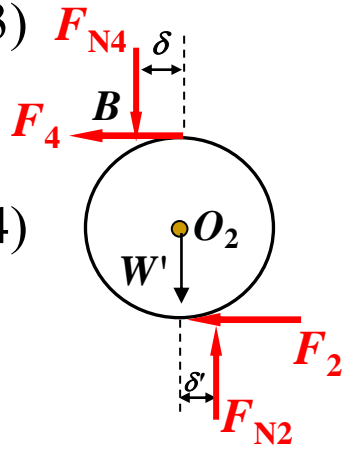
$$\sum F_y = 0 \quad -W - 2W' + F_{N1} + F_{N2} = 0 \qquad (2)$$

2. Take the rolling wood on the left for research



$$\sum M_A(F_i) = 0 \quad F_{N1}(\delta + \delta') - 2F_1 r - W'\delta = 0 \qquad (3)$$

3. Take the rolling wood on the right for research



$$\sum M_B(F_i) = 0 \quad F_{N2}(\delta + \delta') - 2F_2 r - W'\delta = 0 \qquad (4)$$

Solving simultaneous equations （1） （2） （3） （4）：

$$F=\frac{W\left(\delta+\delta'\right)+2W'\delta'}{2r}$$

Discussion： if W=1000kN，W'=0，δ=0.05cm，

δ'=0.20cm，r=12.5cm。solved：F=10kN。

(2) if $\delta=\delta'=0$ 时 $F=0$ 。it is equivalent to placing a heavy object on an ideal smooth surface

End of chapter 4